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Agentic Crystallization v2.1

A Bounded Self-Regulating Procedure Controller with Independent Renewal and Delayed Drift Tests

Research release: 2.1.0
Status: Executable synthetic testbed, conditional statistical guarantees, adversarial failure cases.
Scientific boundary: Not a physical crystal, not an established phase transition, not a production-grade self-sustaining agent, and not an integration with Pegasus.

Full mathematics · Algorithm card · Experiments and limitations · Frozen design · Historical v2.0

Abstract

An adaptive procedure system that retains previously successful computational structures faces two different challenges: learning which structures improve performance and learning when historical evidence is no longer applicable. Versions 1.0–2.0 of Agentic Crystallization developed synthetic higher-order procedure interactions, value-aware information acquisition, sequential certification, and a finite procedure-discovery/retention loop. Their governance boundary remained dependent on externally declared environmental-shift allowances. Version 2.1 adds an independent, retrospective two-sample degradation monitor and formal memory states for fresh certification, renewal and retirement. A lifetime split risk budget uses one-sided Hoeffding bounds for current-block procedure promotion and separately for detection of deterioration in labeled post-deployment outcomes. The experiment compares six policies across eight seeded synthetic environments, including gradual change, abrupt inversion, recurring tasks, unlearnable concepts, missing labels, and deliberately forged feedback. On 13,824 policy-epoch decisions from 24 held-out worlds, the controller reactivates memory usefully, detects major abrupt changes, and reduces repeated proposal-search cost. However, it misses gradual shifts and fails under misleading feedback; periodic pre-block validation dominates under rapid change. An eight-world independent replication preserves these qualitative results. The system is a bounded **self-regulating research prototype**, not evidence of unconditional future correctness or indefinite self-sustaining intelligence.

1. Research objective and the word “crystal”

Physical crystallization arises from matter interacting under specific physical constraints. Digital structures do not spontaneously organize into adaptive systems simply because they encode in-

formation. A computational procedure becomes more useful over time only if it is selected by a meaningful evaluation process, retained with provenance, and challenged again by new observations. We use *crystal* as a precise **engineering analogy** for a persistent but revisable network of useful procedure combinations, not as a claim about thermodynamics or consciousness.

Our target is an auditable controller that can repeatedly perform:

1. Construct candidate procedures from a finite permitted algebra.
2. Rank candidates using exploratory feedback isolated from subsequent validation.
3. Certify a positive current-block gain over an explicit baseline.
4. Retain the validated structure as a *candidate for future reuse*, never as permanent authority.
5. Measure subsequent outcome deterioration and withdraw continued use when the evidence is sufficiently strong.
6. Expire stale entries, retire repeatedly degraded entries, and recover through fresh, independent certification.
7. Bound computational expenditure and durably record all decisions.

Version 2.0 implemented most of this loop, but trusted an externally asserted source-to-deployment shift allowance. Version 2.1 changes the evaluation contract: when independent *current-block* task labels are available, use fresh within-block validation and spend a separate error budget on a retrospective drift test. This avoids pretending a past certificate covers an unobserved future distribution. It does **not** eliminate the need for the environment to supply honest and representative evidence.

2. Minimal mathematical substrate

Let $\mathcal{P}$ be a finite set of permitted procedures and $a_0 \in \mathcal{P}$ a baseline. For each task observation $Z = (X, Y)$ and candidate a, measure the paired improvement

$$D_a(Z) = \mathbf{1}\{a(X) = Y\} - \mathbf{1}\{a_0(X) = Y\} \in [-1, 1].$$

At epoch e, the environment may have different distributions $P_{e,0}$, $P_{e,1}$, and $P_{e,2}$ in three consecutive task blocks. Define $\mu_{e,b}(a) = \mathbb{E}_{P_{e,b}} D_a(Z)$. We deliberately do **not** assume $\mu_{e,0} = \mu_{e,1} = \mu_{e,2}$.

The retention state M_e is a bounded mapping from procedure identifiers to the status and dated evidence record. The complete operational state is

$$\mathcal{C}_e = (\mathcal{P}, M_e, \mathsf{Cert}_e, \mathsf{Monitor}_e, B_e, L_e).$$

The final two components are available synthetic computation credits and an integrity-checked append-only receipt ledger. An external stream of tasks and labeled outcomes drives the state transition; the system does not produce energy or validated facts from nothing.

Historical interaction result

For independent fair binary features, a planted three-way XOR target has no empty-baseline one- or two-feature predictive advantage, yet the complete triple achieves perfect accuracy. The corresponding three-way Möbius interaction in Bayes accuracy is $+\frac{1}{2}$. This exact algebra is preserved from earlier releases; it demonstrates that nonadditive *planted* informational utility is possible. It does not prove emergent phase criticality or a general tendency of all digital information to crystallize.

3. Theorem A: fresh current-block promotion

Select a single candidate a **before** reading its validation sample. Let n_j range over the four predeclared checkpoints $\{64, 128, 256, 512\}$. Assign a unique certification slot $t = 3(e-1) + b + 1$ and spend

$$\alpha_t^{\text{prom}} = \frac{0.025}{t(t+1)}.$$

Define the one-sided lower confidence bound

$$\boxed{L_{t,j}(a) = \widehat{\mu}_{t,j}(a) - \sqrt{\frac{2\log(4/\alpha_t^{\text{prom}})}{n_j}}}.$$

Promote for the **named current block** only if $L_{t,j}(a) > \tau$ with $\tau = 0.025$. Otherwise return HOLD and deploy the baseline. The candidate may have been selected adaptively using old receipts, scouting and discovery observations, but not the validation sample.

Conditional guarantee. If the validation differences are independent, identically distributed and bounded in $[-1, 1]$ conditional on the past, and represent the current block's task law, then

$$\boxed{\Pr(\text{at least one false positive current-block promotion over a controller lifetime}) \leq 0.025.}$$

Proof sketch. Hoeffding's one-sided probability of an upward mean error exceeding the radius at a checkpoint is at most $\alpha_t^{\text{prom}}/4$. Unioning four checkpoints yields α_t^{prom} for one slot; summing the telescoping $1/[t(t+1)]$ allocations yields at most 0.025 for all possible slots. Complete derivation: MATHEMATICS.md.

This is a standard concentration bound used carefully in a versioned controller. It **does not certify future blocks** without additional premises. A harmful change after current-block validation is not a "false certificate" for the original block; it is a separate deployment-exposure failure.

4. Theorem B: retrospective loss-of-value monitoring

A candidate validated early in an epoch is measured again **after middle-block task outcomes arrive**. Keep a 512-sample fresh reference batch and an independently observed 768-outcome monitoring batch. Let their means be $\widehat{\mu}_r$ and $\widehat{\mu}_c$.

Reserve 0.025 separately for monitoring, with index $s = 3(e-1)+2$ and $\beta_s = 0.025/[s(s+1)]$. A two-sample one-sided confidence bound for the change in expected gain is

$$\boxed{U_e^{\text{drop}} = \widehat{\mu}_c - \widehat{\mu}_r + \sqrt{\frac{2\log(2/\beta_s)}{512}} + \sqrt{\frac{2\log(2/\beta_s)}{768}}.}$$

The controller raises `ALARM` when $U_e^{\text{drop}} < 0$. Under independent representative labeled batches, this bounds the lifetime chance of declaring statistically significant deterioration when the expected gain did not deteriorate by **at most 0.025**. This monitor does not assess absolute harm unless a separate absolute-risk test is performed; degradation and harm are distinct propositions.

Chronological caveat: The label sample comes from actions already taken in the middle block. Even if the test were perfectly powerful, detection follows exposure and cannot retroactively protect those tasks. Nor does a `CLEAR` result establish stationarity: it means only that no statistically significant drop was detected with the available sample and allocated risk.

5. State transitions and governance

A procedure's historic status can be `RETAINED`, `NEEDS_RENEWAL`, or `RETIRED`. A new epoch converts retained structures into proposed-only candidates. Fresh source-period validation is required to make any active promotion. A deterioration alarm marks the current procedure as needing renewal; two alarms (or prolonged staleness) retire it from memory-based nominations. If permitted, the controller then explores and independently certifies a replacement for block two. On missing monitoring labels it must use the baseline in the final block.

The state-machine rules prevent unauthorized procedure execution and reject malformed candidate IDs. Valid procedure compositions are restricted to XORs of at most three of eight Boolean inputs, plus baseline zero. The actor cannot synthesize arbitrary code, modify its own source, or change authority boundaries.

Each epoch begins with a conservative cost reserve. After observable task performance and evaluation charges, the ledger commits

$$\boxed{B_{e+1} = B_e - C_e + R_e \geq 0, \qquad C_e \leq B_e.}$$

External tasks pay the synthetic reward $R_e \geq 0$. Nonnegative credit balance is not perpetual economic or physical sustainability. Receipts are SHA-256-linked and replayable after interruption; this verifies integrity of the provided file, not its authenticity against a malicious owner who rewrites the entire chain.

6. Frozen evaluation protocol

The first exploratory cohort used seeds 20000–20003. After a small receipt-validation hardening change, the final source hashes and untouched holdout seeds were recorded in PREREGISTRATION_STAGE_M.json. The final holdout used seeds 51000–51023, 12 epochs per world, eight regimes and six policies (**13,824 policy-epochs**). A separate eight-world replication used seeds 81000–81007 (**4,608 policy-epochs**), without refitting.

Every epoch contains three deployment blocks of 256, 768 and 512 tasks. The early and later data-generating laws may differ, while fresh independent validation samples match their *own current block* only. The controller is evaluated under stationary repetition, recurrence, abrupt switching, gradual mixture shifts, outcome inversion, four-way unlearnability, missing outcome labels and deliberately misleading labels.

The six policy baselines matter. `memoryless` discovers and certifies each epoch without retained memory. `fixed_reuse` never changes candidate type but does check the same candidate at epoch start. `monitor_hold` stops after the alarm rather than paying for a replacement search. `periodic` acquires fresh labeled validation **before every block**, and therefore represents a more expensive and often safer competitor. `baseline` never leaves the constant-zero predictor.

All numeric scores are **synthetic utility units**, defined by additional correct task predictions minus sampling/computation/memory charges. They are not dollars, runtime milliseconds, physical joules, real-world task success probabilities, or measures of autonomy.

7. Independent results

Environment	Crystal utility/epoch	Memoryless	Periodic	Key result
Stationary	753.19	751.87	740.15	Memory saves some search; fixed reuse slightly cheaper

Environment	Crystal utility/epoch	Memoryless	Periodic	Key result
Recurrent	**721.08**	699.82	**741.23**	216 memory procedures freshly recertified
Abrupt shift	**364.12**	110.63	**740.07**	288 alarms; periodic pre-block validator wins
Gradual shift	511.15	509.83	446.63	**288 deteriorations missed**
Inverted outcomes	**−283.93**	−526.49	**+86.72**	288 alarms, **288 harmful pre-alarm blocks**
Unlearnable	208.47	211.53	188.18	No observed false promotions of prohibited four-way parity
Missing feedback	111.61	110.39	**739.41**	Late-block HOLD, loss of opportunity
Misleading feedback	**−525.17**	−526.49	**+86.72**	**0 alarms, 576 harmful blocks**

The crystal's recurrent-task gain over memoryless is **+21.265** synthetic units per epoch (descriptive paired world-bootstrap 95% interval **[+20.807,+21.720]**). Against periodic certification under abrupt change, its deficit is **−375.951** (interval **[−377.560,−374.315]**). These intervals resample the 24 world seeds; they are descriptive statistics, not theorem-backed guarantees of generalization.

Independent eight-world replication

Qualitative conclusions persist in the separate replication cohort: recurrent crystal mean 719.65 versus memoryless 697.90, abrupt crystal 365.37 versus periodic 741.37, and misleading-feedback crystal −525.06 versus periodic +87.50. The same synthetic world generator produces both cohorts, so this is not out-of-family transfer.

8. An identifiability boundary for every "self-regulating" system

Suppose two possible environments yield exactly the same observations through the decision point, but one environment rewards continued reuse while the other penalizes it after an unobserved switch. Any algorithm that bases its decision on the shared history must have the same decision distribution in both worlds. No amount of source-only computation can determine the future law from information the environment has not revealed.

Stage M exhibits the result directly: the `misleading_feedback` condition inverts real task outcomes but reports monitoring labels drawn from the prior unchanged rule. The crystal cannot distinguish the change from stationarity and incurs 576 harmful blocks without an alarm. If feedback is absent, HOLD is a defensible conservative action; it is not proof of accurate environmental understanding.

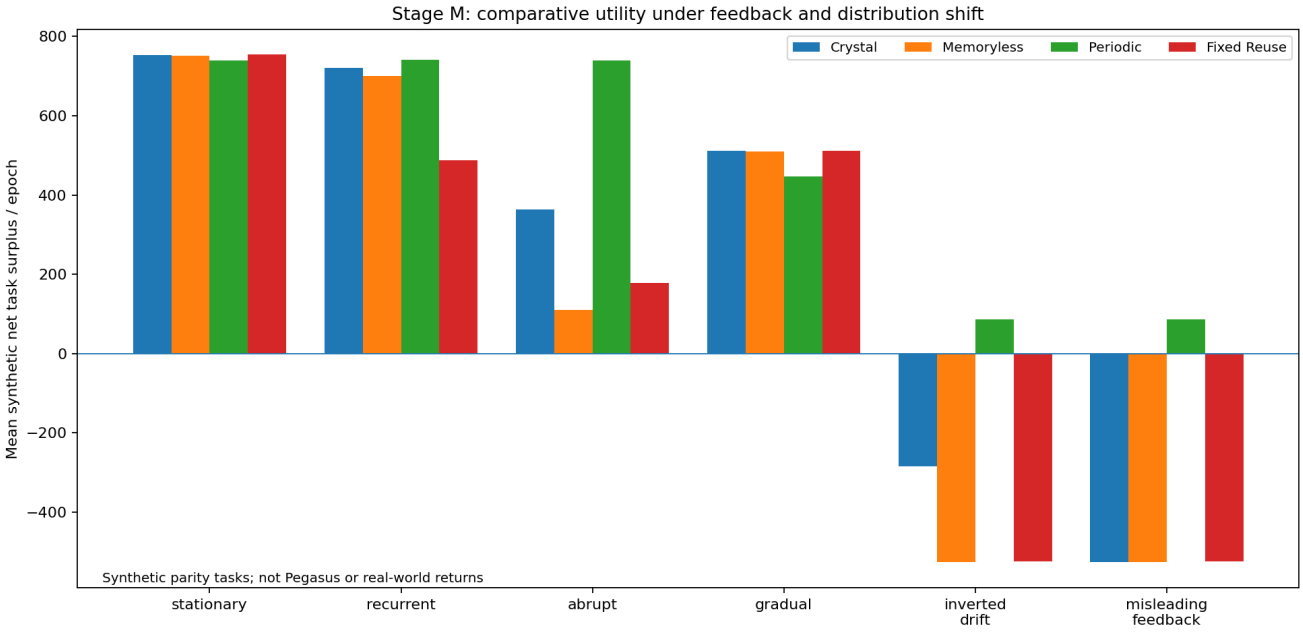


Figure 1: Policy utilities across held-out environments

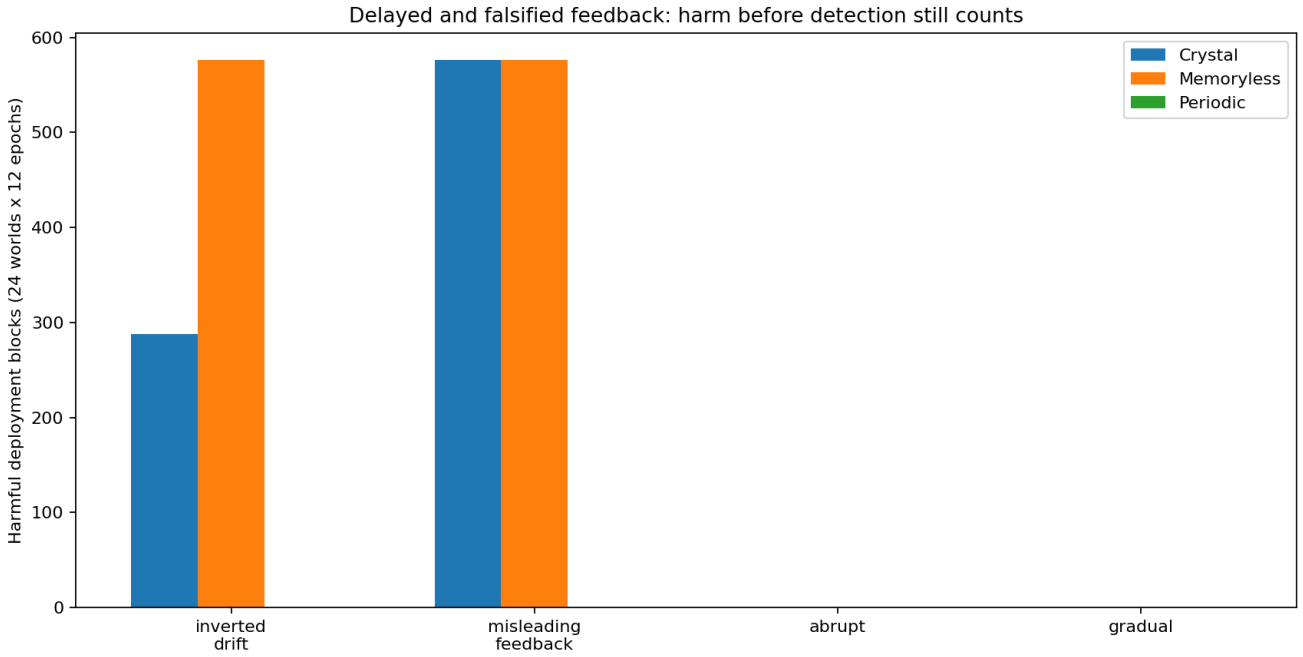


Figure 2: Harmful deployment blocks show the cost of delayed or false feedback

This forces an engineering requirement: a deployed agent needs **trusted, independent measurement of current outcomes** or must treat past certifications as provisional. More elaborate coherence scores do not solve missing information.

9. Counterarguments and limitations

1. **Unlearned environment representations.** The algorithm searches a fixed, permitted XOR family; it does not discover a new representation language or improve the search algorithm itself.
2. **No proved phase transition.** Neither finite-size critical scaling nor a universal threshold was tested. The "crystal" remains an analogy to selective retention.
3. **Measurement quality is assumed.** Delayed and adversarially corrupted feedback provide observed failure cases. The monitor cannot authenticate external task labels.
4. **Current-block scope.** Mathematical promotion certificates cover the block sampled for validation, not subsequent unobserved blocks.
5. **Finite resource model.** An initial credit reserve and externally rewarded tasks permit a finite run. Neither energy closure nor indefinite economic surplus is demonstrated.
6. **No Pegasus result.** No live LLM, CodeFlow harness, real-world agent skill, or authentic policy promotion was used.
7. **Uncertain generalization.** All performance outcomes depend on task prevalence, planted XOR structure, observation sample sizes and synthetic cost ratios. An eight-world replication cannot establish deployment robustness.
8. **Receipt authenticity.** A checksum chain is not a signature; a sufficiently privileged adversary can rewrite every record and recompute hashes.

10. Conclusion and v2.2 research hypothesis

Version 2.1 achieves an auditable, bounded *feedback-regulated procedural retention* loop on controlled tasks. It integrates fresh certification, remembered candidates, retrospective degradation alarms, renewal/retirement, finite resources, and restartable receipt records. Importantly, it shows the mechanism's strong limit: future environmental change is not observable until trustworthy data reveal it, and the cost of discovering that change may exceed the cost of occasional advance recertification.

A falsifiable v2.2 question follows: can the controller **select between delayed outcome monitoring, early sentinel evaluations, and full periodic recertification** based on measured costs and uncertainty, without oracle knowledge of task phase or feedback integrity? Test this against sparse labels, deceptive channels, distribution shift, and resource constraints with a new untouched holdout. No fixed improvement, self-sustaining behavior, or new law should be asserted without that evidence.

Audit resources: Exact proofs, Algorithm Card, Experimental report, Frozen protocol, Raw main holdout, Raw replication, Prior full whitepaper v2.0.